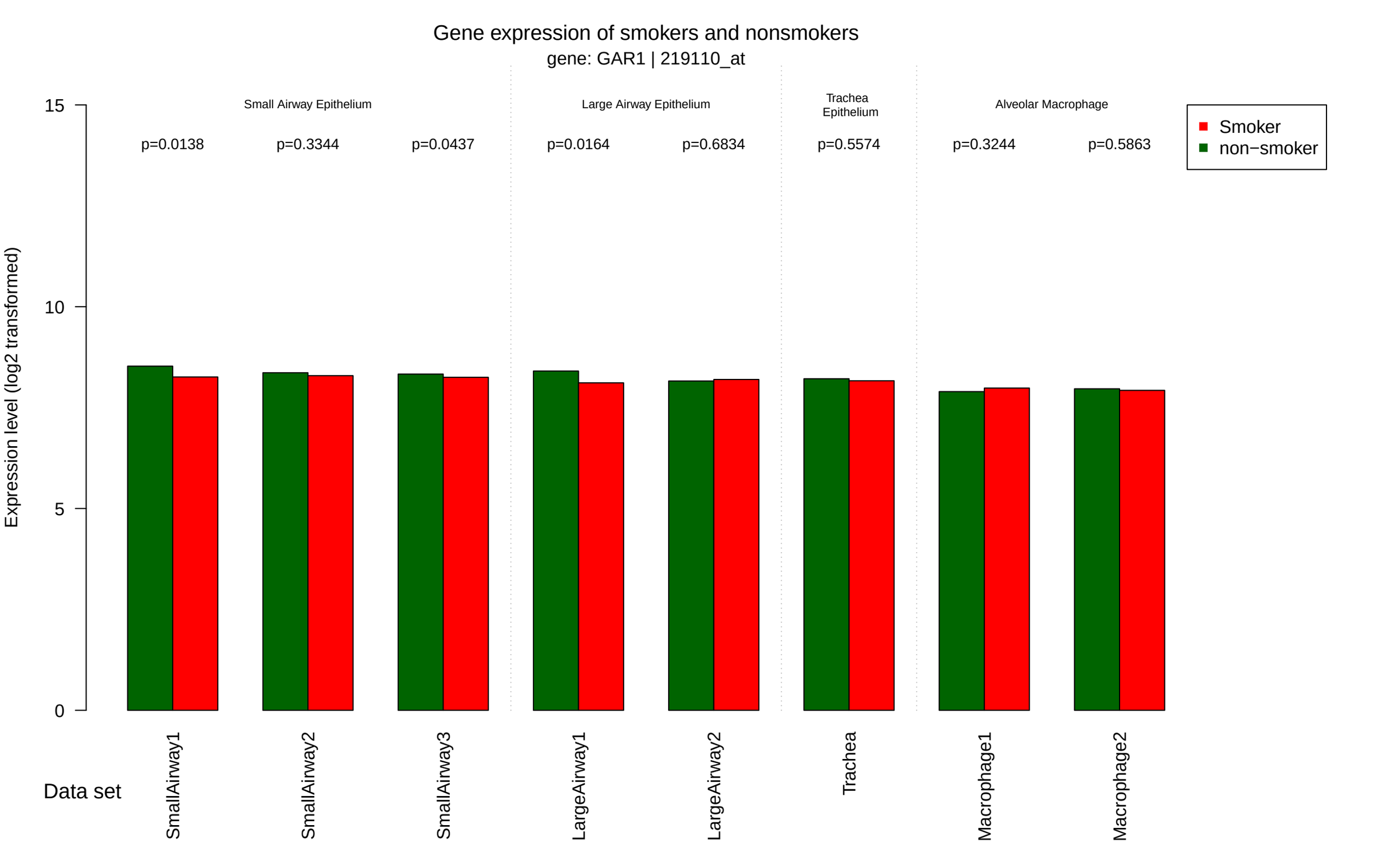
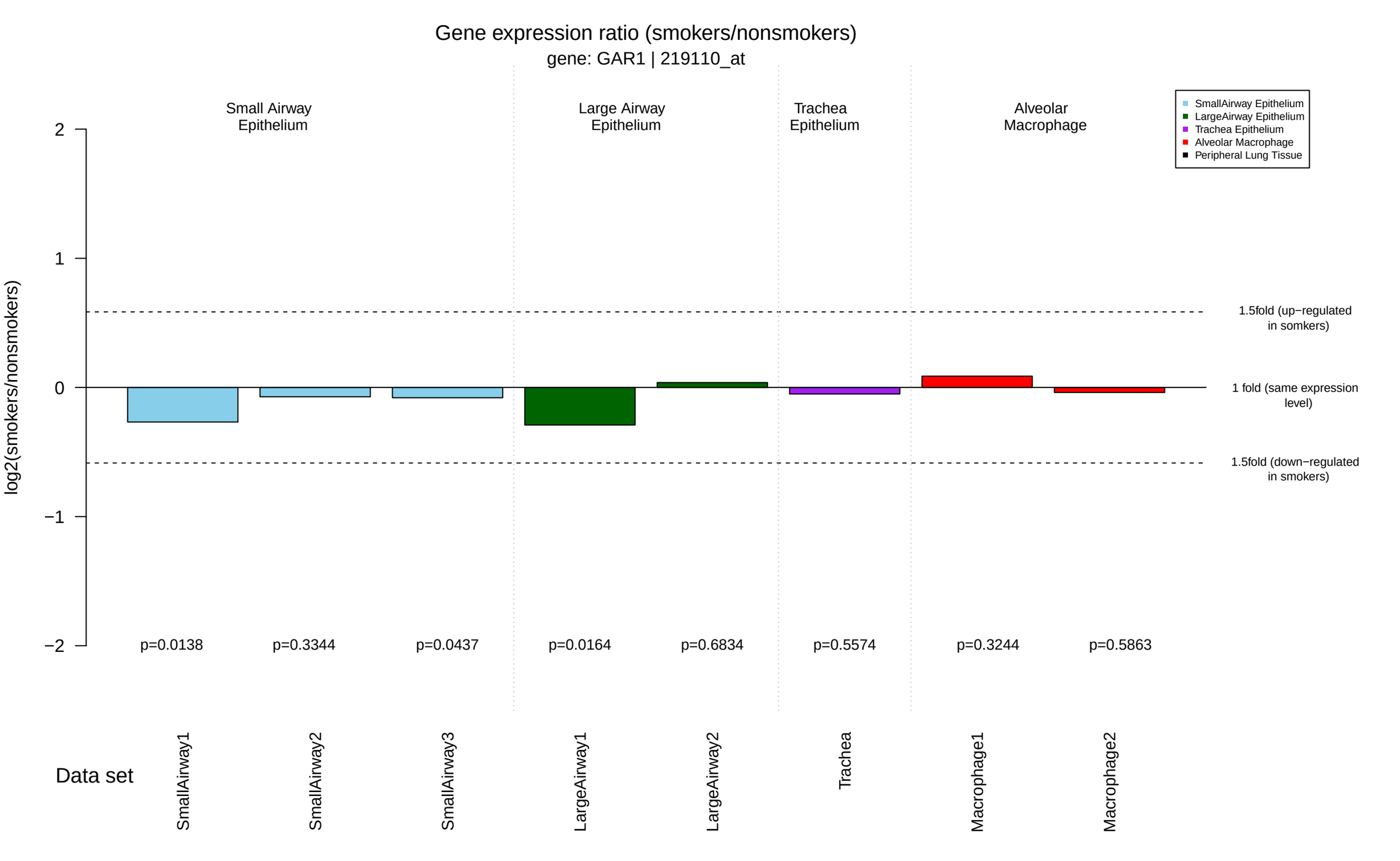
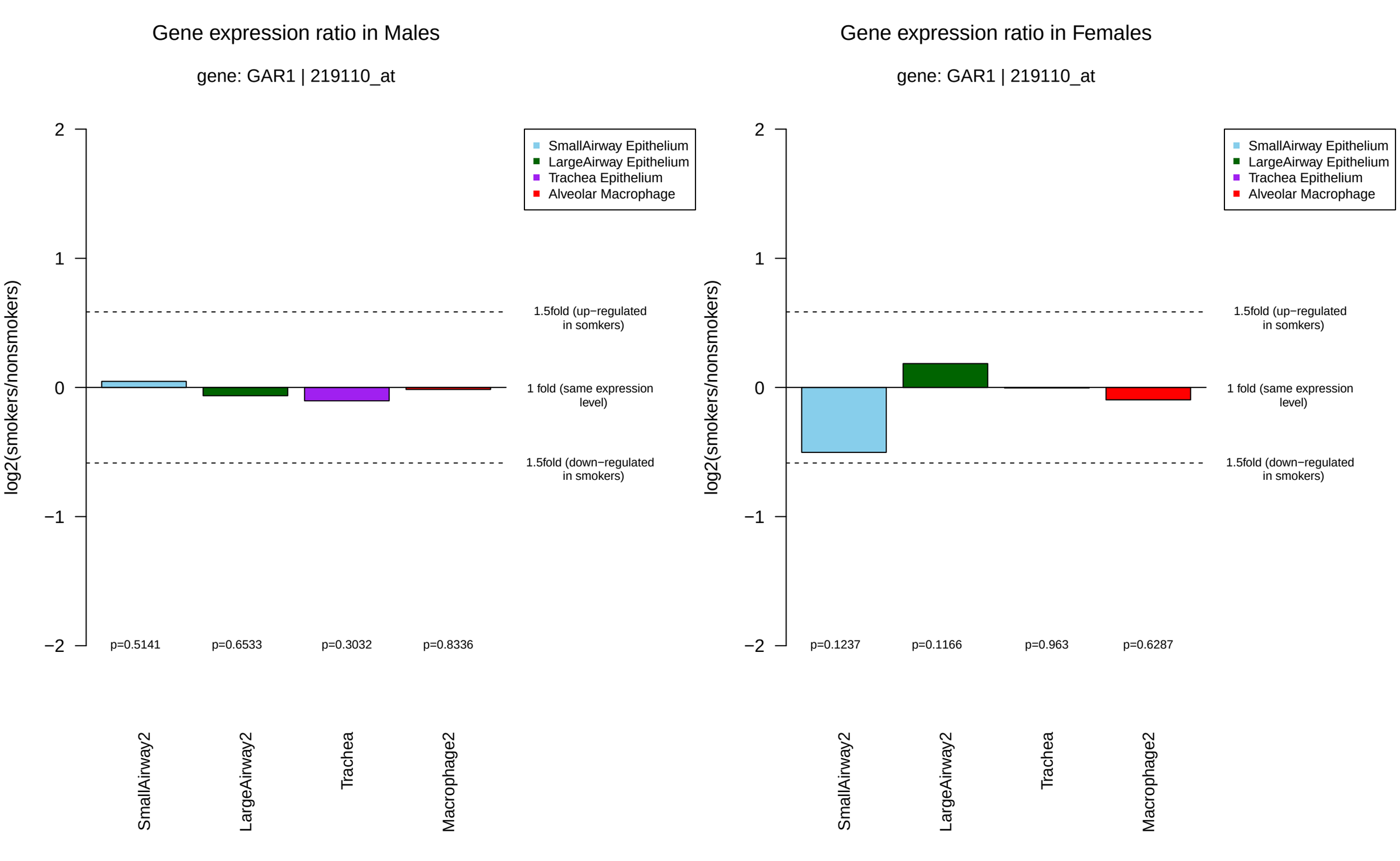
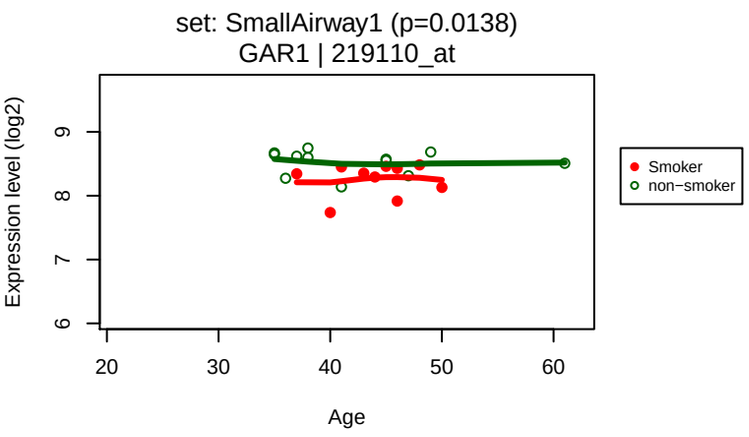
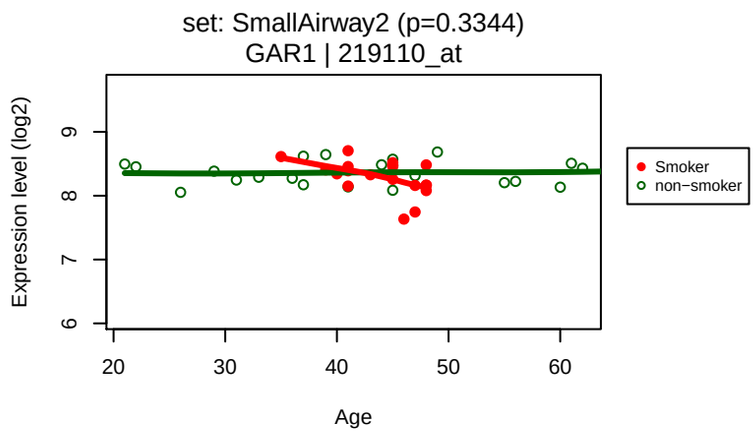
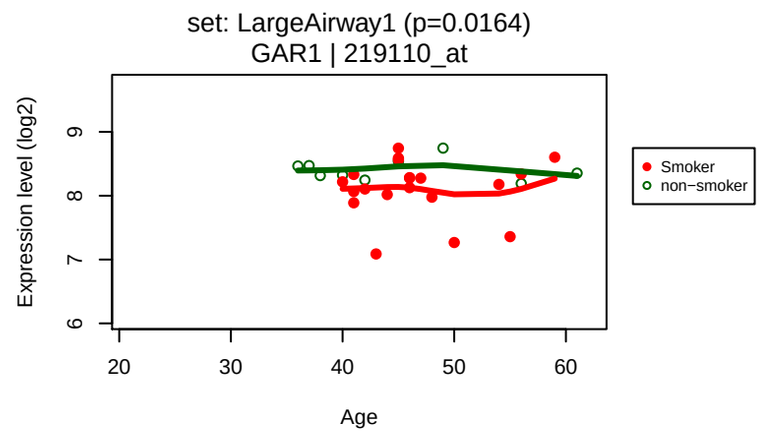

Gene expression of smokers and nonsmokers
gene: GAR1 | 219110_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0138
p=0.3344
p=0.0437
p=0.0164
p=0.6834
p=0.5574
p=0.3244
p=0.5863
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

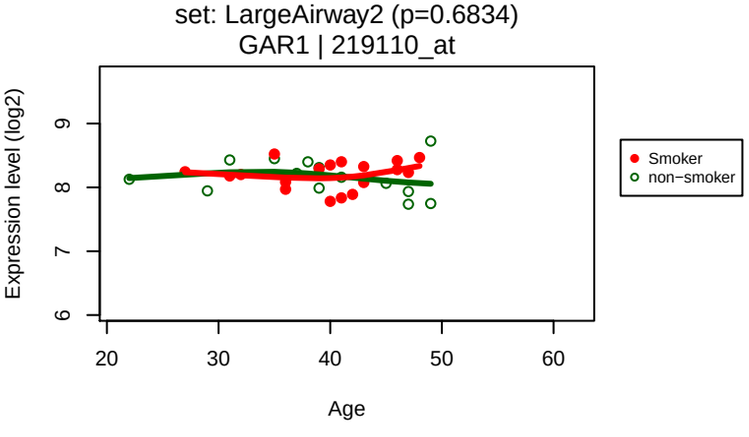

Gene expression ratio (smokers/nonsmokers)
gene: GAR1 | 219110_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0138
p=0.3344
p=0.0437
p=0.0164
p=0.6834
p=0.5574
p=0.3244
p=0.5863
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

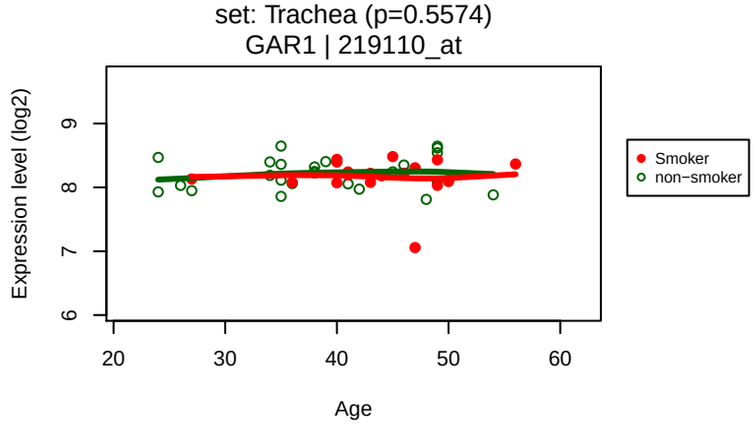

Gene expression ratio in Males
gene: GAR1 | 219110_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5141
p=0.6533
p=0.3032
p=0.8336
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAR1 | 219110_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1237
p=0.1166
p=0.963
p=0.6287
SmallAirway2
LargeAirway2
Trachea
Macrophage2

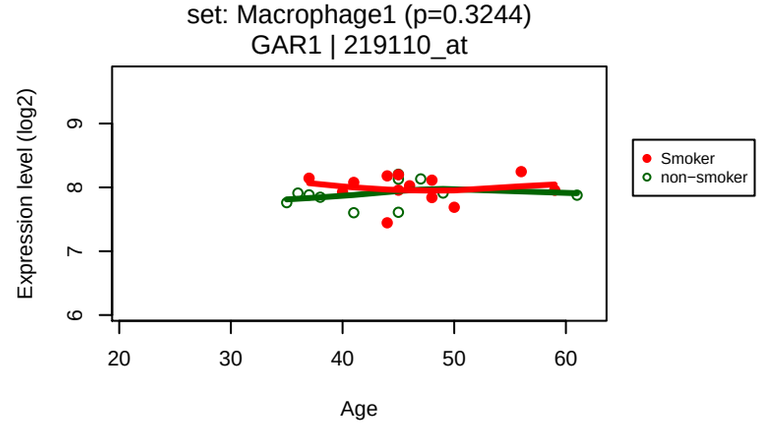

set: SmallAirway1 (p=0.0138)
GAR1 | 219110_at
Expression level (log2)
Age
Smoker
non−smoker

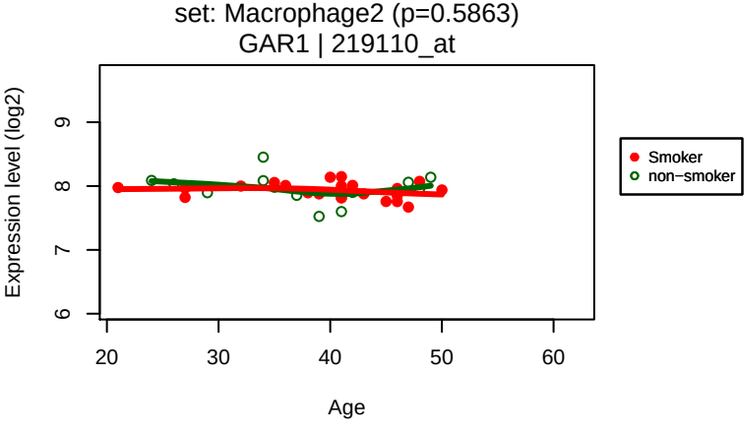

set: SmallAirway2 (p=0.3344)
GAR1 | 219110_at
Expression level (log2)
Age
Smoker
non−smoker

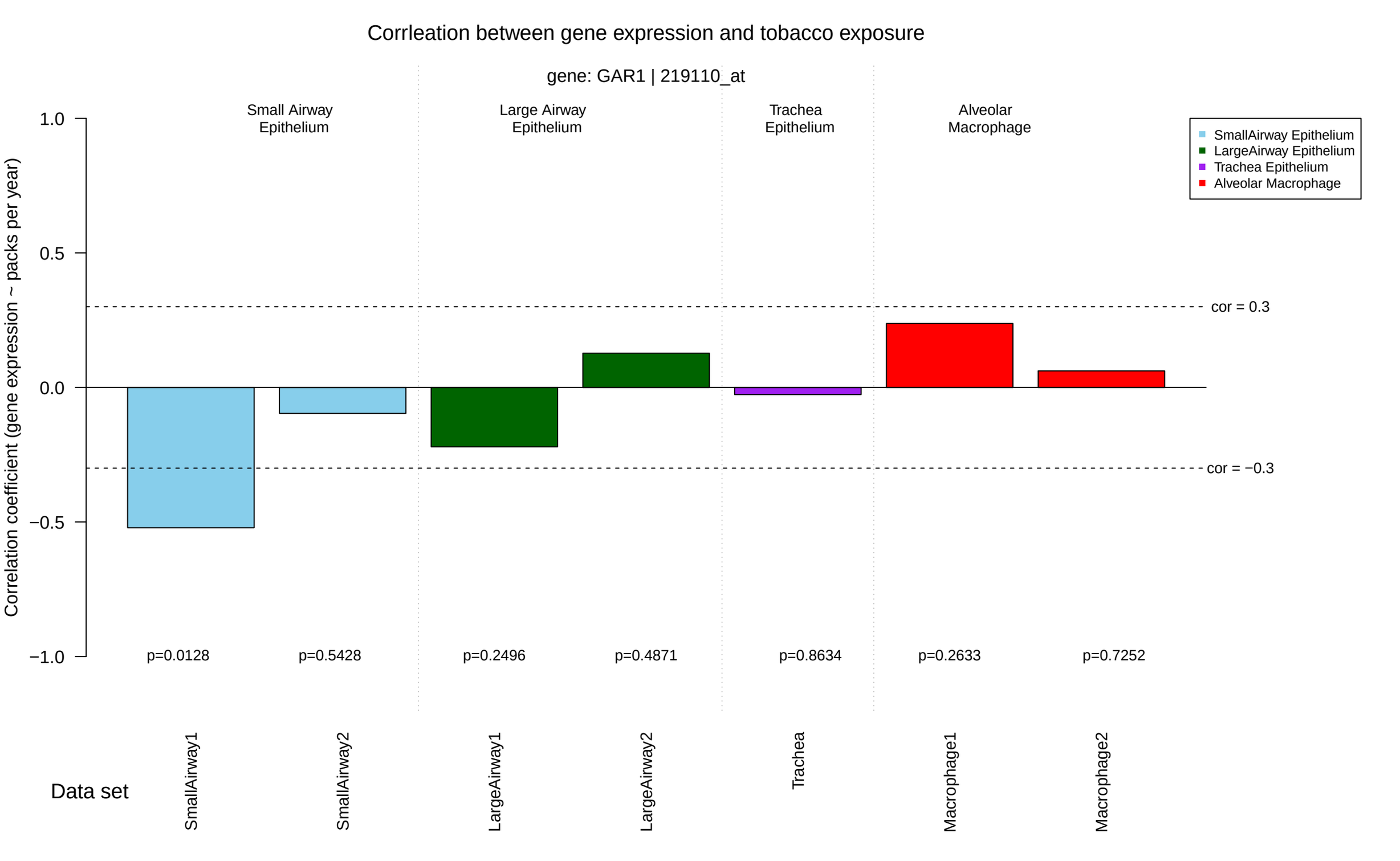

set: LargeAirway1 (p=0.0164)
GAR1 | 219110_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6834)
GAR1 | 219110_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5574)
GAR1 | 219110_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3244)
GAR1 | 219110_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5863)
GAR1 | 219110_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAR1 | 219110_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0128
p=0.5428
p=0.2496
p=0.4871
p=0.8634
p=0.2633
p=0.7252
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2